

Youth Electronic Media Use and Family Conflict: Bidirectional Associations Across Early Adolescence

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The present study examined the longitudinal, bidirectional associations between early adolescents' electronic media use and family conflict across 3 years using data from the ABCD study (T1; $N = 11,787$). Findings indicated that more overall electronic media use was related to later increases in family conflict, whereas the reverse association was not supported. Further analyses examined associations by unique types of electronic media. Social electronic media use at T1 in girls was related to less family conflict at T2, yet more family conflict at later timepoints. These findings suggest that more overall electronic media use promotes later conflict throughout early adolescence and that girls' use of social types of media may be especially disruptive in families as they progress through early adolescence.

Keywords: electronic media use, adolescence, screen time, family conflict, social media

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Electronic media use (EMU) has increased year over year since 2011 and American youth now collectively spend more time using electronic media than with any other leisure activity. This rapid proliferation of digital activities has transformed the way youth engage in social relationships during the transition to adolescence, including in the family. Simultaneously, the family environment continues to exert profound influence on youth well-being and behavior throughout adolescence. Within this complex interplay of electronic media and family dynamics, it is increasingly critical to understand the potential

bidirectional influences between factors that shape developmental trajectories.

The relations among EMU and family relationships have been a subject of considerable interest and concern among developmental scholars, policymakers, and parents alike. On one hand, electronic media has been hailed as a transformative force with the potential to enhance communication and connectivity (Allen et al., 2014). On the other hand, it has raised concerns about the effects of excessive screen time on quality of family interactions (Mesch, 2006b) and family disputes over parental controls of screen time (Halpin et al., 2021),

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The data that support the findings of this study are available to qualified researchers with a valid data use contract in the National Institute of Mental Health Data Archive (NDA) at <https://nda.nih.gov>, NDA Collection No. 2573, <https://nda.nih.gov/study.html?id=2313>. Data and ideas for the present study have not been previously disseminated. The present study utilized data from the Adolescent Brain Cognitive Development (ABCD) Study, which is a publicly available data set. The ABCD study was conducted in compliance with the ethical standards set forth by the relevant institutional review boards at each participating site. Informed consent and assent were obtained from all participants and their legal guardians. The current analysis was conducted under data use agreement, and all procedures adhered to the ethical guidelines and principles of the Helsinki Declaration and other relevant ethical standards. Informed consent and assent were obtained from all ABCD Study participants and/or their legal guardians at the time of enrollment. All consenting participants have further given consent for data sharing for secondary data analysis. Data used in the preparation of this article were obtained from the ABCD study (<https://abcdstudy.org>), held in the NDA. This is a multisite, longitudinal study designed to recruit more than 10,000 children aged 9–10 and follow them over 10 years into early adulthood. ABCD consortium investigators designed and implemented the study and/or provided data but did not necessarily participate in the analysis or writing of this report. This article reflects the views of the authors and may not reflect the opinions or views of the National Institutes of Health or ABCD consortium investigators. The ABCD data repository grows and

changes over time. The ABCD data used in this report came from <https://nda.nih.gov/study.html?id=2313>. The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Cory Carvalho played a lead role in conceptualization, data curation, formal analysis, methodology, project administration, software, supervision, visualization, writing—original draft, and writing—review and editing. Nyantri Ravindran played a supporting role in conceptualization, methodology, and writing—review and editing. Kalsea J. Koss played a lead role in methodology and a supporting role in conceptualization, formal analysis, writing—original draft, and writing—review and editing.

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which may lead to higher levels of family conflict. At the same time, family conflict represents a significant challenge in the lives of adolescents, as it has been consistently associated with a range of negative outcomes, including emotional distress (Koss et al., 2017), academic underachievement (Dotterer et al., 2008), and other problem behaviors (Cummings et al., 2015).

Indeed, early empirical evidence suggests that excessive EMU by youth is related to dysfunction within family relationships (Carvalho et al., 2015), including higher levels of family conflict (Mesch, 2006a). However, these studies are often crosssectional, limiting our ability to disentangle direction of effects. Alternatively, higher levels of family conflict may increase young people's tendency to use electronic media as an avoidant coping strategy. It is plausible that associations among EMU and family conflict are bidirectional and age dependent across early adolescence. Overall, studies have yet to investigate these associations longitudinally and bidirectionally, which represents a substantial gap in the literature. Further, it may be that EMU effects across development may depend on the type of media youth engage with (e.g., social media, television, video sharing sites [e.g., YouTube], video games, and texting). However, this lack of specificity has been identified as a gap in the EMU effect literature (Kaye et al., 2020).

We aim to address these gaps in the present study by leveraging a large, nationally representative data set of the U.S. youth. First, we examine the bidirectional longitudinal associations between total youth EMU and family conflict annually across four waves of data. Next, we test the associations between family conflict and specific types of electronic media. By examining the nuanced relations between these two influential factors over time, this research aims to inform understanding of youth development and provide insights for parents, educators, and policymakers who are navigating the ever-evolving landscape of technology and family dynamics.

Electronic Media Usage Patterns in Early Adolescence

There has been a significant and continuous rise in the use of electronic media in recent years among early adolescent youth (Rideout et al., 2022; Twenge et al., 2019). This surge in digital engagement has been driven by the proliferation of screen technologies, which have become increasingly accessible and affordable (Twenge et al., 2019). The introduction to electronic media typically occurs early in a child's life (Paudel et al., 2017), and its prevalence steadily increases throughout childhood. By middle childhood (ages 5–8), parents report that their children spend an average of 3 hr per day immersed in screen media (Rideout & Robb, 2020). During this stage, electronic media activities primarily consist of gaming, television viewing, and streaming shows and takes place on various devices such as tablets, desktop computers, parents' smartphones, and gaming consoles (Rideout & Robb, 2020).

As children enter early adolescence, their daily EMU escalates to approximately 6 hr per day, excluding time spent on educational activities and homework (Rideout et al., 2022). Adolescence is characterized by escalations in gaming and television consumption as well as the emergence of social media and texting with one in five preteens reporting daily social media usage with higher rates in girls than boys (Rideout et al., 2022). Given this change in the way many youth spend their time within the electronic media landscape, it becomes imperative to investigate potential associations

between different types of electronic media and family relationship dynamics.

Family Conflict in Preadolescence

Family conflict is a family-level behavior characterized by disagreement and discord among members of a familial unit that typically includes parents and their children (Cummings et al., 2015). Low and infrequent levels of family conflict that include minor disagreements and criticisms is an aspect of normative family interactions (A. W. Collins et al., 2004). At these levels, youth are exposed to manageable stress and conflict resolution strategies that can facilitate the development of prosocial skills (Dishion et al., 2012; Thompson & Meyer, 2007). However, frequent and elevated levels of conflict involving high tempers, acrimony, and even physical altercations between family members is a known risk factor for negative socioemotional development (Cummings & Schatz, 2012).

The transition from childhood to adolescence has been identified as a period of elevated family conflict (W. A. Collins et al., 1997). This is partly due to substantive neural and physiological developmental changes and their downstream psychosocial effects that usher in increased need for independence (Steinberg, 2007). For example, youth at this age begin to seek increasing levels of autonomy and reliance on peers for support. Such changes can result in a temporary disruption of previously stable patterns of family interactions until families can reorganize relationship patterns (Branje, 2018). This makes families with early adolescent youth especially vulnerable to conflict, and it is critical to identify factors that contribute to added stress on relationships in families during this time in development.

Effect of Youth EMU on Family Conflict

Excessive youth EMU is one factor that may contribute to family conflict during adolescence. As youth increasingly spend their time and attention with screens, the quality of face-to-face interactions within the family may diminish (McDaniel, 2014; Tammissalo & Rotkirch, 2022). Reduced opportunities for meaningful communication, shared activities, and emotional bonding can erode the sense of closeness that is essential for healthy family functioning (Mesch, 2006b). Consequently, this attenuation of family closeness can contribute to dynamics in which misunderstandings and tensions may thrive, exacerbating parent-child conflicts and straining familial relationships (Mesch, 2006a).

Additionally, disputes over screen time rules and content consumption can act as a catalyst for family conflict. As youth immerse themselves in the digital realm, disagreements can arise within families over appropriate screen time limits, content choices, and the timing of device use (Evans et al., 2011; Reid Chassiakos et al., 2016). These disputes can evolve into persistent sources of tension within the family, leading to heightened levels of parent-child conflict (Francis et al., 2021). The struggle to establish and enforce boundaries related to screen time can further erode communication (Francis et al., 2021), undermining the family's ability to resolve conflicts effectively. Thus, youth screen time may serve as a contentious issue, capable of fostering family discord when disagreements surrounding its management remain unresolved.

Effect of Family Conflict on Youth Electronic Media Use

Family conflict is a plausible driver of EMU among youth. Research indicates that the family environment significantly influences the leisure activities of youth and can affect patterns of EMU (Livingstone et al., 2015). Youth exposed to higher levels of family conflict may turn to electronic media as a means of coping with interpersonal stress (Squires et al., 2021). Supporting this idea, empirical evidence suggests that some individuals use electronic media to regulate emotions and manage stress (Wolfers & Schneider, 2020). Additionally, as youth progress from childhood to adolescence, they tend to employ more active coping methods, such as media consumption (Eschenbeck et al., 2018). Moreover, peer support on social media for dealing with family conflict becomes increasingly available with youth age (Fredrick et al., 2022). However, research exploring the impact of family conflict on EMU during early adolescence remains limited.

Specificity of Electronic Media

Electronic media is a comprehensive term encompassing various digital activities, including video games, video viewing, and social forms, all of which occur on electronic devices. These activities share common traits, such as high convenience and maximizing user engagement through rewarding and stimulating content. However, there are noteworthy distinctions in the ways youth engage with each type of media, potentially contributing to distinct associations with family conflict. This lack of specificity in this area of research has been identified as a limitation in the existing electronic media effect literature (Kaye et al., 2020).

Television and video sharing sites, such as YouTube, are passive forms of electronic media where users predominantly engage in *video viewing* of content. Typically, exposure to such content commences early in childhood, with usage levels gradually increasing (Vandewater et al., 2007). Texting and social media are inherently *social* forms of electronic media. Texting involves the instantaneous exchange of written messages among friends and family on various devices and applications (Bagot et al., 2022). Similarly, social media applications (e.g., Facebook, Instagram) facilitate social interaction by enabling users to view and share various forms of content, including videos, photos, and thoughts. *Video games* offer a source of active engagement in play and leisure activities. Video games can be played individually and in multiplayer formats with or against peers. Moreover, the prevalence of multiplayer video game use increases with age (Rideout & Robb, 2020; Rideout et al., 2022).

Several factors suggest that socially oriented electronic media may exhibit a stronger association with family conflict than other forms in early adolescence. First, whereas video games and video viewing are common activities throughout early to late childhood (Rideout & Robb, 2020), youth often begin using social forms of electronic media during early adolescence (Pew Research Center, 2022). This introduction of novel features into household dynamics may lead to conflicts as families navigate establishing new boundaries (Reid Chassiakos et al., 2016). Additionally, youth may be influenced emotionally by online social interactions (e.g., cyberbullying, conflict with peers), affecting their behavior at home. In the other direction, youth experiencing higher levels of family conflict may be particularly drawn to social media and texting in search of peer social

support and as relief from family discord. Last, whereas other forms of media can be enjoyed as a family (e.g., watching movies and sports, multiplayer video games), social media and texting with peers often exclude those in the immediate physical environment. Nevertheless, to date, there is a dearth of research exploring the bidirectional effects between specific electronic media types and family conflict.

The Present Study

Early adolescence is marked by elevated levels of both family conflict and EMU. Thus, it is germane to understand the nature of their relations across early adolescent development and the current lack of studies in this area represents a significant gap in research to date. First, we will test the bidirectional effects between youth overall EMU and family conflict across four annual time points. We hypothesize that there will be bidirectional effects such that higher levels of EMU will lead to more family conflict and higher levels of family conflict will lead to more EMU. Next, we will separately explore the bidirectional associations between family conflict and specific types of media, including video viewing, video games, and social. We hypothesize that the bidirectional associations between EMU and family conflict will be strongest in the social EMU models. In addition, given differences in the way boys and girls use some types of EMU (Rideout et al., 2022), we explore whether there are gender differences across the models tested.

Method

We used data from the Adolescent Brain Cognitive Development (ABCD) Study to address study aims. The ABCD Study recruited over 11,500 adolescents aged 9–10 across 21 sites beginning in 2015 with follow-up data collection planned every 6 months for a total of 10 years (Garavan et al., 2018). This sample size was determined to be adequate for sufficient power to detect small effects accounting for an anticipated 10% attrition across the duration of the study. The ABCD Study implemented minimal exclusion criteria to achieve a representative sample. Exclusion criteria included (a) child's lack of English proficiency; (b) significant sensory, neurological, medical, or intellectual impairments that could hinder compliance with protocols; (c) children unable to complete magnetic resonance image scanning at baseline. Overall, data collected consist of neuroimaging, cognitive tasks, and survey data. Data collection sites were selected based on research expertise and neuroimaging equipment availability and were thus largely restricted to urban areas, yet catchment areas were set to a 50-mile radius intentionally to extend into rural populations (see Garavan et al., 2018, for review of sample design considerations). The ABCD study was conducted in compliance with the ethical standards set forth by the relevant institutional review boards at each participating site. The data that support the findings of this study are available to qualified researchers with a valid data use contract in the National Institute of Mental Health Data Archive at <https://nda.nih.gov>. This study was not preregistered. We report how sample size, all data exclusions, all manipulations, and all measures in the study were determined.

Participants

For the present study, survey data were used from baseline (T1: $N = 11,876$; $M_{\text{age}} = 9.9$), 1-year follow-up (T2: $N = 11,219$; $M_{\text{age}} = 10.9$), 2-year follow-up (T3: $N = 100,972$; $M_{\text{age}} = 12.0$), and 3-year follow-up (T4: $N = 10,335$; $M_{\text{age}} = 12.9$). The sample aimed to be a representative sample of U.S. adolescents (White = 52.0%, Hispanic = 20.3%, Black = 15.0%, Asian = 2.1%, Other = 10.5%; Female = 47.8%). For income, 31.2% of children were from families with an annual income of less than \$50,000 compared to the national average of 42.2% and 40.8% reported annual income of \$100,000 or higher compared with the national average of 27.3%. Moreover, 73.5% were from married, two-parent households compared to the national average of 66.1% (Heeringa & Berglund, 2020).

Measures

Electronic Media Use

Youth reported on their own daily EMU via the Screen Time Questionnaire (Bagot et al., 2022) at T1, T2, T3, and T4. In an ABCD cohort substudy ($N = 67$) that compared youth with parent-reported screen time against an objective, passive sensing app, youth were more accurate when reporting screen time than parents (youth: $r = .49$, $p < .001$; parent: $r = .10$, $p = .43$; Wade et al., 2021); thus, we rely on youth self-reports in this study. For types of EMU, youth were asked about time spent using specific types of EMU. Specific types of EMU include television and online streaming (at T1, T2, and T3), texting and social media, and video games (at T1, T2, T3, and T4). For each type, youth were separately asked the following multiple-choice questions about weekday and weekend day hours (i.e., two questions in total) "On a typical [WEEKEND DAY/WEEKDAY], how many hours do you?" Specifically, participants were asked how often they "Watch TV shows or movies?" "Watch videos (Such as YouTube)?" "Play video games on a computer, console, phone, or other device?" "Text on a phone, tablet, or computer (e.g., GChat, Whatsapp, etc.)?" and "Visit social networking sites like Facebook, Twitter, Instagram, etc.?" Youth were asked in multiple-choice format how many hours (i.e., 1, 2, 3, or 4) and minutes (i.e., 0, 15, 30, and 45) they were engaged with each EMU type. Weekday and weekend totals were highly correlated across EMU types ($r_{\text{min}} = .68$, $p < .001$; $r_{\text{max}} = .78$, $p < .001$). Thus, a weighted mean for specific types of media use was calculated for each type. Composite scores were generated for social (i.e., social media + texting) and video-viewing types (i.e., streaming + television). Notably, data for T4 video viewing were not available. Also, at T4, the video game EMU survey item changed from a single item to a composite of time spent playing single-player and multiplayer video games.

The ABCD Study provided total daily EMU measures in addition to the daily averages for each type. For T1 and T2, the provided total daily EMU was a composite of all EMU types. We similarly calculated total daily EMU at T3 and T4 composites of all EMU types, respectively. Correlations in years 2 and 3 between the screen time total item and screen time composite scores were high ($r_2 = .6$, $p < .001$; $r_3 = .7$, $p < .001$). Youth reported more screen time overall when reporting individual types than when asked singularly about their overall screen use. Media multitasking is common in youth (e.g., watching tv while on social media) which likely contributes to

the observation that the sum of media types is greater than the overall totals when reported singularly. EMU data were winsorized to three standard deviations above and below the mean.

Family Conflict

Youth report of family conflict was measured using the nine-item conflict dimension of the Family Environment Scale (Moos, 1990; Moos & Moos, 1994) at T1, T2, T3, and T4. Youth responded True or False to each item (Zucker et al., 2018). The total number of True statements were summed (possible range 0–9). Items include verbal and physical aspects of family conflict. Example items for this survey were: "Family members often criticize each other," "We fight a lot in our family," and "Family members sometimes hit each other." Reliability for the original validation study was good ($\alpha = .75$; Moos, 1990) and in the current sample was adequate at each time point ($\alpha_1 = .68$, $\alpha_2 = .67$, $\alpha_3 = .64$, $\alpha_4 = .68$).

Covariates

Youth sex and family income were included as covariates. In addition, analyses were stratified by site ID to account for any location bias resulting from the multisite design of the ABCD Study.

Analytic Plan

First, descriptive statistics and bivariate correlations were analyzed. Second, a random-intercept crosslagged panel model was used to examine bidirectional effects between youth total EMU and family conflict (Hamaker et al., 2015). This method isolates effects related to within-person variability from between-person associations by accounting for individual trait-like components (Mulder & Hamaker, 2021). With this approach, one can test whether the between-person trait-like components are related by correlating the random intercepts. In addition, the within-person autoregressive effects (e.g., stability) and crossconstruct effects between study variables can be modeled as well. Using this approach, crosslagged effects across four waves (age 9–10 to age 12–13) of total youth EMU and family conflict were analyzed. Additionally, to assess whether effects varied by type of EMU, we analyzed the bidirectional effects of specific types of EMU and family conflict. Three separate models assessing specific types of EMU were analyzed (i.e., video game EMU, social EMU, and video-viewing EMU). Three waves were available for video game and video-viewing EMU and four waves were available for social EMU. Missing data increased over time (T2: 5.6%–5.7%, T3: 7.8%–7.9%, T4: 13.2%). Analyses of attrition patterns revealed that boys ($F = 5.95$, $p = .015$), youth from lower income families ($F = 111.21$, $p < .001$), those reporting higher levels of baseline EMU ($F = 10.00$, $p = .002$), and individuals experiencing greater baseline family conflict ($F = 5.74$, $p = .017$) were slightly more likely to drop out after T1. All study analyses were conducted using Mplus Version 8.3 using full information maximum likelihood with robust standard errors estimation because it produces unbiased parameter estimates with nonnormal data (Yuan & Bentler, 2000).

To test whether the strength of the associations among EMU and family conflict vary across time during adolescence, a series of models were fit. The selection of the best fitting models was determined via a multiple-step process based on Satorra–Bentler

scaled chi-square difference tests which are recommended for comparing models using maximum likelihood with robust standard errors estimation (Satorra & Bentler, 2001). First, stability and crosslagged parameters were constrained to be equal in M1. Then, crosslagged and stability parameters were allowed to vary across time in M2 and M3, respectively. Next, all parameters were allowed to vary across time in M4. If M2, M3, and M4 models fit all improved significantly over M1, then M4 was selected. If model fit improved significantly for either M2 or M3, but not both, then the better fitting model was selected.

To test whether associations varied among boys and girls, the selected model from previous steps was then used to compare model fit between gender-based multigroup models to determine whether a gender-based analysis was appropriate. The sequential process outlined in the preceding steps was repeated, only this time by constraining and freeing stability and crosslagged parameters across boys and girls. First, stability and crosslagged parameters were constrained to be equal across boys and girls in M5. Then, crosslagged and stability parameters were allowed to vary across boys and girls in M6 and M7, respectively. Next, all parameters were allowed to vary across gender in M8. If M5 was determined to be the best fitting multigroup model, then the best fitting model from the previous step was selected (i.e., M1, M2, M3, or M4). If a better fitting model was found than M5, then that superior model was used in analysis. Model fit of the best fitting models was evaluated for acceptability along established fit index cutoffs (comparative fit index [CFI] > .95, Tucker–Lewis index [TLI] > .95, root-mean-square error of approximation [RMSEA] < .08, standardized root-mean-square residual [SRMR] < .08). For whole sample models (i.e., without gender differences), income and youth sex effects on family conflict and EMU at each timepoint were added as control variables after the best fitting models were selected. When the best fitting model featured gender differences, only income was included as a control variable. Modification indices were reviewed if the addition of covariates weakened model fit. The Benjamini–Hochberg false discovery rate technique was also used to account for multiple tests.

Transparency and Openness

The data used in this study are derived from the ABCD Study, which is publicly available to qualified researchers. Access to the ABCD data can be obtained through the National Institute of Mental Health Data Archive at <https://nda.nih.gov/abcd>. Researchers interested in using the data must complete the requisite data use certifications and adhere to the National Institute of Mental Health Data Archive's data use agreements. This study was not preregistered.

Results

Descriptive Statistics and Correlations

Means, standard deviations, and correlations between study variables are shown in Table 1. Repeated-measure analysis of variance results suggest that there were significant changes in EMU and family conflict across time (see Supplemental Table 1). Specifically, all EMU types increased across early adolescence while family conflict showed an early decrease and a later increase across the study period. All types of EMU were positively associated

with family conflict across all time points. Higher family income was associated with lower levels of family conflict and EMU across all types. Moreover, across all timepoints, correlations revealed boys used significantly more viewing ($r_{\min} = .05, p < .001$), video games ($r_{\min} = .30, p < .001$), and total electronic media ($r_{\min} = .06, p < .001$) while girls used more social media ($r_{\min} = -.12, p < .001$).

Random-Intercept Crosslagged Panel Model

Total EMU and Family Conflict

For total EMU, the best fitting model featured constrained crosslagged and freed autoregressive parameters suggesting the strength of the crosslagged associations remained similar in magnitude across adolescence while the strength of the autoregressive parameters varied by child age (M3; see Supplemental Table 2). Additionally, there was no support for gender differences in the total EMU model. Model fit was very good, CFI = .99; TLI = .97; RMSEA = .03; SRMR = .02; $\chi^2(26) = 316.01, p < .001$. Parameter estimates for this model are displayed in Figure 1. There was a significant between-person positive association between family conflict and total EMU ($b = 0.88; \beta = .33; SE = .05; p < .001, p\text{-adj} < .001; 95\% \text{ CI } [0.79, 0.96]$). For within-person effects, higher levels of total screen time at time t predicted more family conflict 1 year later at each time point $t + 1$ ($b = 0.02; \beta = .04; SE = .01; p = .002; p\text{-adj} = .003, 95\% \text{ CI } [0.01, 0.03]$), however family conflict at time t did not predict screen time at the next time point ($b = 0.00; \beta = .01; SE = .01; p = .501, p\text{-adj} = .623; 95\% \text{ CI } [-0.03, 0.04]$).

Video-Viewing EMU

The best fitting model for video-viewing EMU exhibited no gender differences and equality constraints on crosslagged and autoregressive parameters suggesting no variations in the strength of associations across development (M1; see Supplemental Table 3). Model fit was excellent, CFI = .99; TLI = .99; RMSEA = .011; SRMR = .01; $\chi^2(14) = 41.63, p < .001$. Parameter estimates for this model are displayed in Figure 2. There was a significant between-person positive association between family conflict and video viewing ($b = 0.49; \beta = .36; SE = .03; p < .001, p\text{-adj} < .001; 95\% \text{ CI } [0.43, 0.55]$). For within-person pathways, video-viewing EMU did not predict later family conflict ($b = 0.02; \beta = .02; SE = .02; p = .203, p\text{-adj} = .330, 95\% \text{ CI } [-0.01, 0.05]$) nor did family conflict predict later video-viewing EMU ($b = 0.01; \beta = .01; SE = .01; p = .421, p\text{-adj} = .600; 95\% \text{ CI } [-0.01, 0.03]$). Overall, families with children who used more video-viewing media tended to report more conflict, however there were no within-person effects across early adolescence.

Social EMU

Longitudinal social EMU associations with family conflict can be seen in Figures 3 and 4. The best fitting model exhibited freely estimated parameters across time and did include gender differences suggesting associations varied across early adolescent development and by child gender (M8; see Supplemental Table 4). Model fit was very good, CFI = .99; TLI = .97; RMSEA = .04; SRMR = .02; $\chi^2(20) = 195.91, p < .001$.

Parameter estimates for boys are displayed in Figure 3. There was a significant between-person positive association between family conflict and social EMU ($b = 0.10; \beta = .15; SE = .02; p < .001, p\text{-adj} < .001$;

Table 1
Correlations, Means, and Standard Deviations for Study Variables

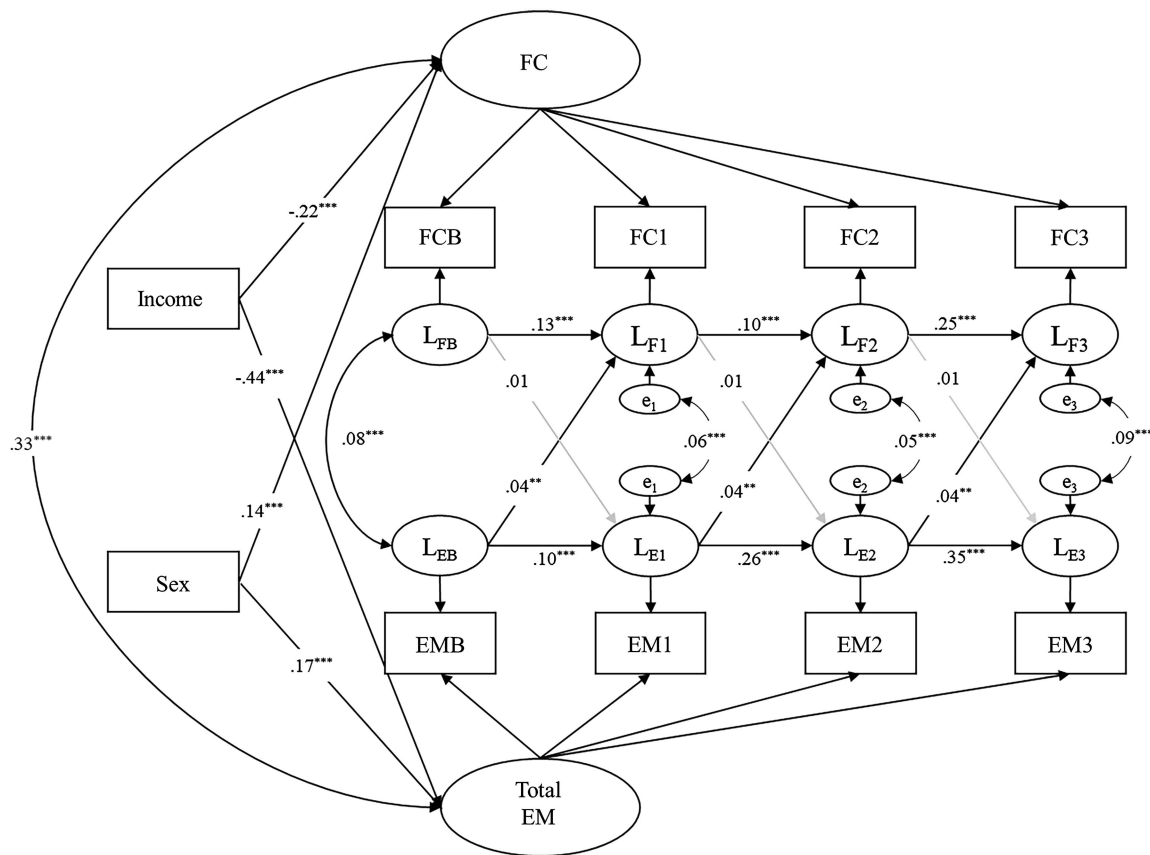
Variable	1	2	3	4	5	6	7	8	9	10
1. Income	—									
2. Sex (1 = male)	.01	—								
3. Total EMU T1	-.29***	.12***	—							
4. Total EMU T2	-.31***	.08***	.55***	—						
5. Total EMU T3	-.32***	.10***	.47***	.56***	—					
6. Total EMU T4	-.32***	.06***	.40***	.47***	.32***	—				
7. Social EMU T1	-.18***	-.07***	.30***	.65***	.25***	.36***	—			
8. Social EMU T2	-.21***	-.12***	.33***	.32***	.32***	.29***	.36***	—		
9. Social EMU T3	-.21***	-.18***	.25***	.36***	.36***	.26***	.31***	.38***	—	
10. Social EMU T4	-.24***	-.22***	.88***	.48***	.42***	.33***	.21***	.24***	.20***	—
11. Viewing EMU T1	-.27***	.05***	.49***	.85***	.47***	.36***	.18***	.36***	.25***	.26***
12. Viewing EMU T2	-.27***	.06***	.41***	.45***	.70***	.35***	.23***	.12***	.08***	.07***
13. Viewing EMU T3	-.18***	.30***	.73***	.39***	.35***	.29***	.14***	.23***	.12***	.10***
14. VG EMU T1	-.27***	.32***	.43***	.72***	.43***	.36***	.10***	.11***	.18***	.11***
15. VG EMU T2	-.20***	.40***	.36***	.41***	.70***	.41***	.08***	.05***	.04***	.04***
16. VG EMU T3	-.14***	.06***	.13***	.13***	.13***	.10***	.07***	.08***	.06***	.05***
17. Family conflict T1	-.16***	.07***	.18***	.19***	.16***	.12***	.09***	.11***	.13***	.10***
18. Family conflict T2	-.14***	.03***	.17***	.18***	.20***	.14***	.09***	.11***	.10***	.13***
19. Family conflict T3	-.09***	.01	.14***	.14***	.15***	.15***	.07***	.08***	.10***	.13***
20. Family conflict T4	11,876	11,876	11,843	11,203	10,918	10,298	11,847	11,204	10,367	10,298
<i>M</i>	7.22	0.52	3.75	3.91	4.83	6.11	0.34	0.61	1.16	1.89
<i>SD</i>	2.42	0.50	2.89	2.95	3.90	4.04	0.82	1.16	1.66	2.07
Variable	11	12	13	14	15	16	17	18	19	20
11. Viewing EMU T1	—									
12. Viewing EMU T2	.50***	—								
13. Viewing EMU T3	.41***	.46***	—							
14. VG EMU T1	.47***	.31***	.27***	—						
15. VG EMU T2	.33***	.49***	.32***	.48***	—					
16. VG EMU T3	.28***	.32***	.45***	.42***	.52***	—				
17. Family conflict T1	.18***	.14***	.13***	.15***	.11***	.12***	—			
18. Family conflict T2	.16***	.18***	.16***	.14***	.16***	.15***	.46***	—		
19. Family conflict T3	.16***	.16***	.18***	.11***	.12***	.13***	.38***	.48***	—	
20. Family conflict T4	.13***	.14***	.13***	.09***	.10***	.10***	.34***	.43***	.54***	—
<i>N</i>	11,841	11,190	10,379	11,843	11,204	10,385	11,844	11,213	10,934	10,321
<i>M</i>	2.24	2.41	2.54	1.01	1.19	1.52	2.04	1.91	1.92	2.10
<i>SD</i>	1.76	1.81	1.94	1.11	1.20	1.46	1.95	1.88	1.83	1.94

Note. EMU = electronic media use; T = timepoint; VG = video game.

** $p < .01$. *** $p < .001$.

Figure 1

Visualization of Statistical Model for Bidirectional Effects Between Family Conflict and Total Electronic Media Use



Note. Comparative fit index = .984, Tucker–Lewis index = .976, root-mean-square error of approximation = .030, standardized root-mean-square residual = .020. For sex: 0 = female, 1 = male. FC = family conflict; EM = electronic media; L_F = within-person family conflict; L_E = within-person electronic media use; e = residuals; B = baseline.
 ** $p < .01$. *** $p < .001$.

95% CI [0.07, 0.13]). Within-person social EMU was not related to later family conflict at any timepoint (t_1-t_2 : $b = 0.06$, $\beta = .02$, $SE = .07$, $p = .435$, 95% CI [-0.08, 0.19]; t_2-t_3 : $b = 0.05$, $\beta = .03$, $SE = .03$, $p = .125$, $p\text{-adj} = .256$, 95% CI [-0.01, 0.11]; t_3-t_4 : $b = -0.01$, $\beta = -.01$, $SE = .02$, $p = .714$, $p\text{-adj} = .884$, 95% CI [-0.05, 0.03]). Moreover, family conflict did not predict later social EMU at any time point (t_1-t_2 : $b = -0.01$, $\beta = -.01$, $SE = .01$, $p = .490$, $p\text{-adj} = .637$, 95% CI [-0.03, 0.01]; t_2-t_3 : $b = -0.02$, $\beta = -.02$, $SE = .02$, $p = .182$, $p\text{-adj} = .314$, 95% CI [-0.05, 0.01]; t_3-t_4 : $b = -0.04$, $\beta = .03$, $SE = .02$, $p = .056$, $p\text{-adj} = .146$, 95% CI [-0.00, 0.07]).

For girls, there was a significant between-person positive association between family conflict and total EMU ($b = 0.20$; $\beta = .24$; $SE = .02$; $p < .001$, $p\text{-adj} < .001$; 95% CI [0.16, 0.24]). Moreover, within-person social EMU was significantly linked to later family conflict across all timepoints (see Figure 4). However, the direction of this effect varied across time. From T1 to T2, more social EMU was related to less family conflict (t_1-t_2 : $b = -0.59$, $\beta = -.18$, $SE = .02$; $p < .001$, $p\text{-adj} < .001$, 95% CI [-0.94, -0.25]) whereas this association was reversed at later timepoints (t_2-t_3 : $b = 0.13$, $\beta = .08$, $SE = .03$, $p < .001$, $p\text{-adj} < .001$, 95% CI [0.06, 0.16]; t_3-t_4 : $b = 0.07$, $\beta = .07$, $SE = .02$, $p < .001$, $p\text{-adj} < .001$, 95% CI [0.03, 0.10]). Family conflict did not predict later

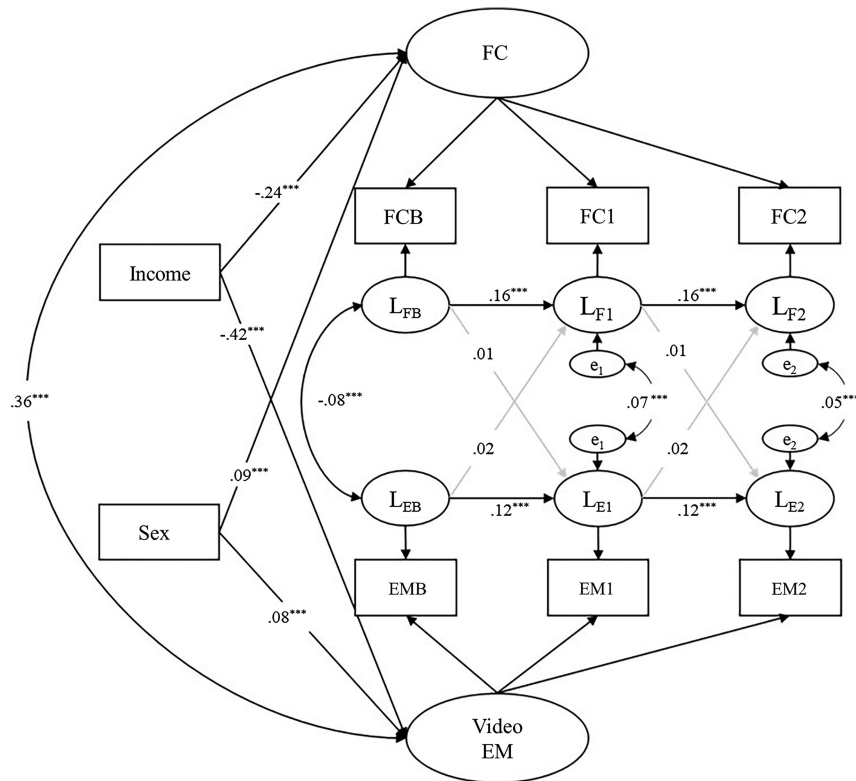
social EMU at any timepoint (t_1-t_2 : $b = -0.05$, $\beta = -.07$, $SE = .02$, $p = .054$, $p\text{-adj} = .146$, 95% CI [-0.09, 0.01]; t_2-t_3 : $b = 0.03$, $\beta = .03$, $SE = .02$, $p = .138$, $p\text{-adj} = .256$, 95% CI [-0.01, 0.07]; t_3-t_4 : $b = 0.04$, $\beta = .03$, $SE = .02$, $p = .051$, $p\text{-adj} = .146$, 95% CI [0.00, 0.09]).

Video Game EMU

For video game EMU, the best fitting model featured no equality constraints on crosslagged or autoregressive parameters and no gender differences suggesting associations varied by development but not by child gender (M4; see Supplemental Table 5). Model fit was excellent, CFI = .99; TLI = .97; RMSEA = .035; SRMR = .021; $\chi^2(19) = 290.5$, $p < .001$. Parameter estimates for this model are displayed in Figure 5. There was a significant between-person positive association between family conflict and total EMU ($b = 0.23$; $\beta = .28$; $SE = .02$; $p < .001$, $p\text{-adj} < .001$; 95% CI [0.73, 0.91]). Video game EMU did not predict later family conflict (t_1-t_2 : $b = 0.04$, $\beta = .03$, $SE = .03$, $p = .089$, $p\text{-adj} = .210$, 95% CI [-0.00, 0.06]; t_2-t_3 : $b = 0.03$, $\beta = .02$, $SE = .02$, $p = .130$, $p\text{-adj} = .256$, 95% CI [-0.01, 0.05]; t_3-t_4 : $b = -0.01$, $\beta = .00$, $SE = .01$, $p = .810$, $p\text{-adj} = .916$, 95% CI [-0.03, 0.02]). However, family conflict was

Figure 2

Visualization of Statistical Model for Bidirectional Effects Between Family Conflict and Video-Viewing Electronic Media Use



Note. Comparative fit index = .998, Tucker-Lewis index = .995, root-mean-square error of approximation = .013, standardized root-mean-square residual = .008. For sex: 0 = female, 1 = male. FC = family conflict; EM = electronic media; L_F = within-person family conflict; L_E = within-person electronic media use; e = residuals; B = baseline.

*** $p < .001$.

linked to more video game EMU later between T2 and T3 (t_2-t_3 : $b = 0.03$, $\beta = .04$, $SE = .01$, $p = .013$, $p\text{-adj} = .048$; 95% CI [0.01, 0.06]), but not at other timepoints (t_1-t_2 : $b = 0.00$, $\beta = .00$, $SE = .01$, $p = .807$, $p\text{-adj} = .916$, 95% CI [-0.03, 0.03]; t_3-t_4 : $b = -0.01$, $\beta = -.02$, $SE = .02$, $p = .226$, $p\text{-adj} = .345$, 95% CI [-0.06, 0.01]).

Sensitivity Analysis

A sensitivity analysis was conducted in which youth behavior problems (internalizing and externalizing types) and parents' screen use behavior around their children (i.e., parental attempt to limit screen use in the presence of their children) were included as covariates by accounting for their effects on the random intercepts of family conflict and EMU in each model. The overall pattern of findings remained similar (e.g., significant effects remained significant, albeit dampened in magnitude). Family conflict was negatively related to parents' screen use ($\beta = -.06$, $p = .001$) and positively youth internalizing ($\beta = .15$, $p < .001$) and externalizing ($\beta = .30$, $p < .001$) problems. Parents' screen behavior was negatively related to youth screen time (total: $\beta = -.14$, $p < .001$; girls social media: $\beta = -.05$, $p = .049$; boys social media: $\beta = -.12$, $p < .001$; video game: $\beta = -.10$, $p < .001$; video viewing: $\beta = -.12$, $p < .001$) and family conflict ($\beta = -.05$, $p = .002$).

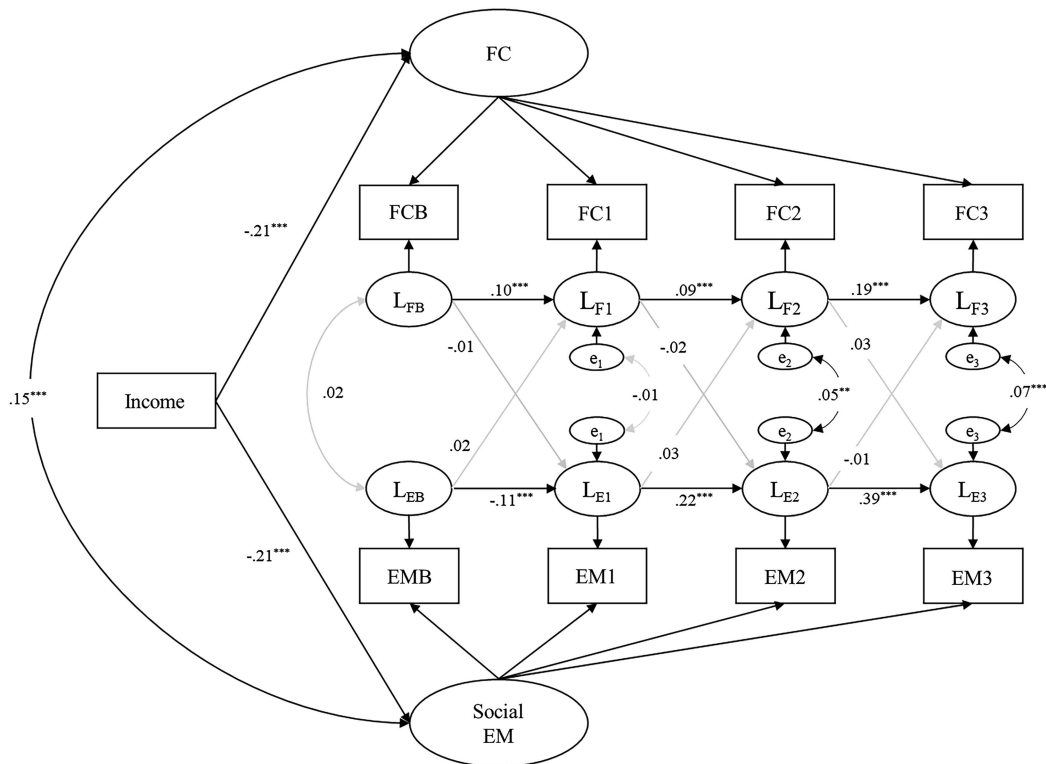
Discussion

The present study examined the between-person and within-person associations among EMU and family conflict. At the between-person level, higher levels of family conflict were associated with higher levels of EMU in all models tested. Then, within-person longitudinal associations between family conflict and youth EMU were also found. First, higher levels of total EMU were associated with higher family conflict levels at subsequent time points. Examining the specific effects by media type revealed that, while most forms were not longitudinally related to family conflict, social EMU was linked to later family conflict in girls. Interestingly, while the direction of this association was negative between T1 and T2, this reversed for later timepoints, and social EMU became a significant risk factor for family conflict. In the reverse direction, family conflict did not generally predict higher EMU, with the exception that youth engaged in more video game use following family conflict between T2 and T3. These findings build on and add nuance to prior research that suggests youth EMU can be a source of conflict within families (Beyens & Buellens, 2017).

Positive between-person associations between family conflict and all EMU variables tested suggest that youth who generally use more

Figure 3

Visualization of Statistical Model for Bidirectional Effects Between Family Conflict and Social Electronic Media Use in Boys



Note. Comparative fit index = .991, Tucker–Lewis index = .966, root-mean-square error of approximation = .038, standardized root-mean-square residual = .015. For sex: 0 = female, 1 = male. FC = family conflict; EM = electronic media; L_F = within-person family conflict; L_E = within-person electronic media use; e = residuals; B = baseline.
 ** $p < .01$. *** $p < .001$.

EMU (i.e., across all time points) come from families characterized by higher levels of conflict. This finding may indicate that processes regarding the causal structure of these effects predate early adolescence resulting in the emergence of associated EMU and family conflict traits by this developmental stage. Although few extant childhood studies have examined this link, some supporting evidence suggests that excessive EMU during childhood is related to more conflict (Oduor et al., 2016), less engagement (Lee & Chae, 2007), and lower levels of cohesion (Yang et al., 2022) in family relationships.

As hypothesized, results indicate that more total EMU is longitudinally related to increased risk of family conflict across early adolescence. Developmentally, youth at this age begin to experience various neurobiological changes that lead to increasing autonomy (Steinberg, 2000) and attendant renegotiation of family relationships over roles, boundaries, and responsibilities (Koepke & Denissen, 2012). This system reorganization contributes to a transient period of more frequent and intense conflicts in some families (Branje, 2018; Ravindran et al., 2020). Placed within this developmental context, our findings suggest that youth EMU aggregated across all types has the potential to exacerbate family conflict during a period often characterized by lowered stability in family relations.

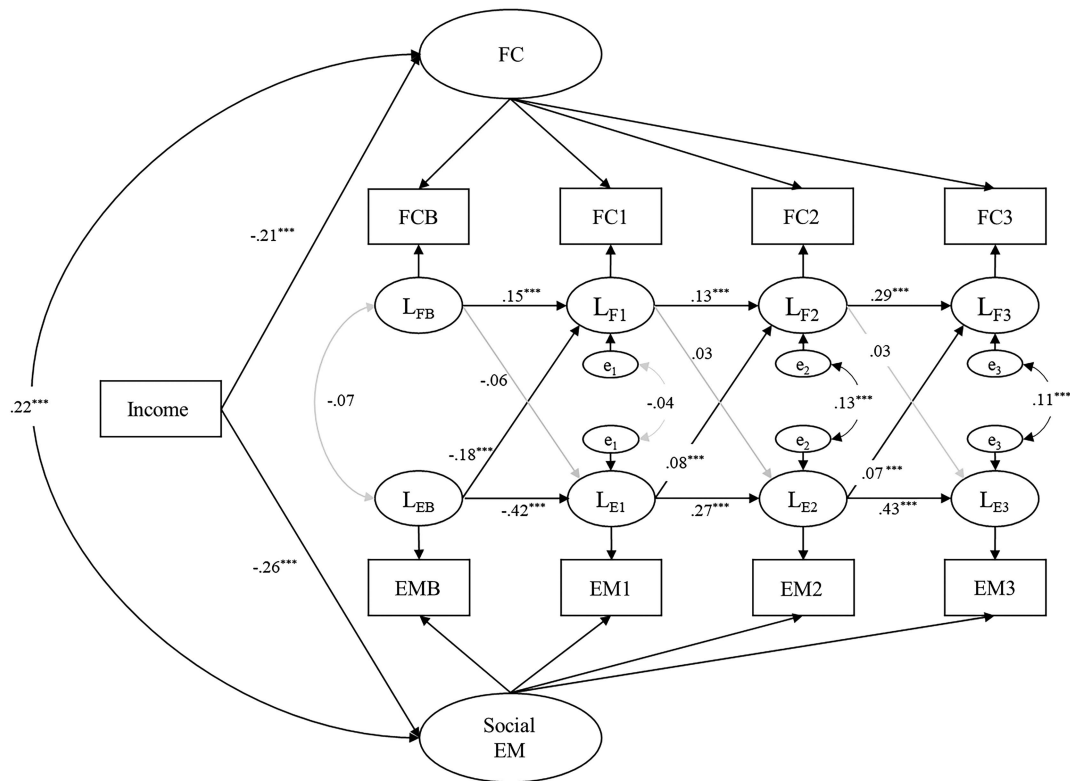
Inspecting the effects of specific types of EMU on family conflict revealed that the social type was the only category that was

linked to later family conflict, in isolation. However, this pattern was unique to girls. This gender divergence may reflect differences in the amount of time spent using social media and texting between boys and girls at this age. Specifically, it has been previously reported that girls spend more time on social media than boys (Rideout et al., 2022; Twenge & Martin, 2020). Moreover, it has been demonstrated that adolescent girls perceive social media to be more socially significant than boys (Leonhardt & Overå, 2021). In support, one meta-analysis found that girls both offer and receive more social support than boys on social media (Tifferet, 2020).

We found that the pattern of associations between girls' social EMU and family conflict was nuanced and the direction of this effect was not uniform across early adolescence. Contrary to hypotheses, T1 (i.e., 9–10 years old) girls social EMU was linked to lower levels of family conflict 1 year later. Social electronic media users at this age would be among the first in their peer group to engage with such activities (Rideout et al., 2022) resulting in few peers to interact with online. Thus, the bulk of their social media and texting behavior may consist of engaging with their parents, older siblings, or extended family. In turn, connecting with parents and siblings via social media and texting can support family relationships (Padilla-Walker et al., 2012) and mitigate conflict. Moreover, without many peers active online it may be

Figure 4

Visualization of Statistical Model for Bidirectional Effects Between Family Conflict and Social Electronic Media Use in Girls



Note. Comparative fit index = .991, Tucker–Lewis index = .966, root-mean-square error of approximation = .038, standardized root-mean-square residual = .015. For sex: 0 = female, 1 = male. FC = family conflict; EM = electronic media; L_F = within-person family conflict; L_E = within-person electronic media use; e = residuals; B = baseline.

****p* < .001.

less challenging for parents and children to reach a consensus as they negotiate social electronic media boundaries. However, no data were collected regarding who youth were engaging in these online social activities with and future studies should test if this explanation is empirically supported.

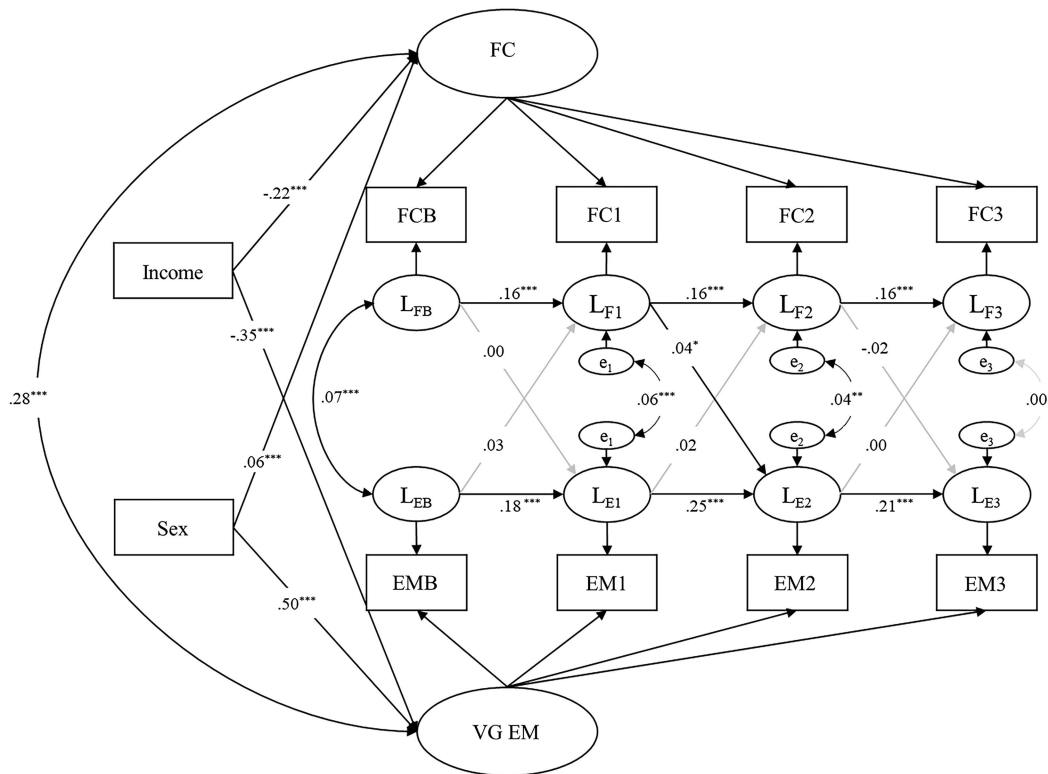
The direction of the effect of social electronic media on family conflict changed after T2 (i.e., 10–11 years old) and it became a significant risk factor for family conflict in girls. It has been found that this is the age that youth typically first engage with social electronic media (Rideout et al., 2022). It may be that as girls increasingly interact with peers online, a growing portion of their social identity development occurs online as well. Thus, logging off in response to family guidelines might then be perceived to be more detrimental to the maintenance of their standing in peer relationships for girls who spend more time engaging socially online. In turn, this could make parents' insistence on abiding by social electronic media rules especially contentious for heavy social media users. This concept can be described in terms of *fear of missing out* (FOMO). FOMO, which has been crosssectionally associated with higher levels of social media use (Zhang et al., 2023), is a documented phenomenon in which individuals feel socially detached from peer groups when absent from online dialogs (Beyens et al., 2016). With FOMO, individuals perceive

that their contemporaries are enjoying meaningful experiences without them, which can result in a high degree of anxiety in some adolescents (Gupta & Sharma, 2021; Oberst et al., 2017). FOMO has been further observed to contribute to youths' difficulty disengaging with and reducing time spent using social electronic media (Fabris et al., 2020), which could result in higher levels of social media use. Notably, youth-reported FOMO was associated with negative parent–adolescent relationship quality in a previous study (Koca & Saatçi, 2022).

Nonsignificant longitudinal within-person associations between video game and video-viewing EMU and later family conflict indicate that there are no changes in how disruptive to the family system these types of media are during early adolescence. This may be because video games and video viewing are normative activities throughout childhood (i.e., prior to the transition to adolescence; Rideout & Robb, 2020) and stable patterns of family interactions have had time to organize around these electronic media types. This explanation is supported by the positive between-person associations that were found. Thus, significant negotiations regarding expectations and boundaries around video games and video viewing may be less common in this age group. It is also possible that families are jointly using television and video games as covieing TV and coplaying video games have been found to

Figure 5

Visualization of Statistical Model for Bidirectional Effects Between Family Conflict and Video Game Electronic Media Use



Note. Comparative fit index = .986, Tucker–Lewis index = .967, root-mean-square error of approximation = .035, standardized root-mean-square residual = .021. For sex: 0 = female, 1 = male. FC = family conflict; EM = electronic media; L_F = within-person family conflict; L_E = within-person electronic media use; e = residuals; B = baseline.
 * $p < .05$. ** $p < .01$. *** $p < .001$.

support family relationships (Hodge et al., 2012; Padilla-Walker et al., 2012).

Contrary to our hypothesis, family conflict did not generally lead to higher EMU. There was one exception to this pattern in which family conflict led to more video game use from T2 to T3. However, this effect was very small, and it was not observed at other time points, necessitating replication. Thus, the evidence found here overall suggests family conflict is not a significant driver of EMU in early adolescence. It may be that using electronic media as a coping strategy in response to the stress of family conflict occurs later in adolescence as youth experience loosened boundaries regarding electronic and mobile devices.

Limitations

Interpretation of these results should be made considering this study's limitations. First, measurements of family conflict and daily EMU were self-reported. For family conflict, observational measures would improve validity. Moreover, instruments that passively monitor electronic devices such as passive viewing apps would improve validity of the EMU data. Second, only youth's EMU was modeled in present study analyses, which hinders our understanding of the broader role of family-level electronic media in

family conflict (Caprì et al., 2021). Future studies should consider parents' and other siblings' EMU behavior in addition to the child. Third, operationalizing EMU as total amount of screen time in developmental research has been criticized for neglecting to consider other aspects of EMU, such as the prevalence of joint versus individual activities, that differ between individuals and families (Kaye et al., 2020). Beyond screen time, future studies should consider the varying functional roles of electronic media in individuals and families. We expand upon the standard use of total screen time and examine associations that vary by type of electronic media. Additionally, family conflict and youths' EMU do not occur in isolation. For example, youth mental health (Formoso et al., 2000), sleep problems (Leonard & Khurana, 2022), and various family stressors (Hartshorne et al., 2021) are known factors associated with both EMU and family conflict. Future studies should disentangle the causal pathways among such constructs to determine mechanistically how processes unfold involving EMU and family conflict across early adolescence. Last, where the present study modeled EMU as a weighted average across weekday and weekend, future studies should test weekday and weekend use separately to determine whether targeted constraints based on part of the week would be more efficient for reducing family conflict. For example, families may have different rules regarding social media use on weekdays versus weekends that

differentially relate to family conflict. Nevertheless, the present study provides support for the directionality among youth EMU and subsequent family conflict during adolescence.

Conclusion

Families are continuously adjusting to an increasingly digital context. Our findings have important implications for the many families who aim to diminish the potential negative impacts of media technologies on healthy family functioning. Here, we found that more total EMU is related to higher family conflict throughout early adolescence. Moreover, girls' social EMU was found to precede higher family conflict as they progress into adolescence, a critical developmental transition. This evidence provides support for moderating adolescent youth's access to social media and texting.

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